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Experimental study of the steady fluid–structure interaction of flexible hydrofoils

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ABSTRACT

This paper presents results from an experimental study of the hydrodynamic and hydroelastic performance of six different flexible hydrofoils of similar geometry; four metal hydrofoils of stainless steel (SS) and aluminum (AL), and two composite hydrofoils of carbon-fiber reinforced plastic (CFRP). The two CFRP hydrofoils had differing layups, one with fibers at 0° and the other at 30° relative to the spanwise axis of the hydrofoil. All hydrofoil models have the same unswept trapezoidal planform of aspect ratio 3.33. Two section profiles were chosen, a standard NACA0009 (Type I) and a modified NACA0009 (Type II) with a thicker trailing edge for improved manufacture of CFRP hydrofoils. Hydrofoils were tested in a water tunnel mounted from a six-component force balance. Forces and deformations were measured at several chord-based Reynolds numbers up to $Re_c = 1.0 \times 10^6$ and incidences beyond stall. Hysteresis, force fluctuations, and the natural frequency of the hydrofoils in air and in water were also investigated. Pre-stall forces on the metal hydrofoils were observed to be Reynolds number dependent for low values but became independent at 0.8×10^6 and greater. Forces on the CFRP hydrofoils presented an increasing or decreasing lift slope for all Re_c depending on the orientation of the carbon unidirectional layers. The change in loading pattern is due to the coupled bend–twist deformation experienced by the hydrofoils under hydrodynamic loading. Forces and deflections in the Type I hydrofoils were observed to be stable up to stall and non-dimensional tip deflections were found to be independent of incidence and Re_c. Type II metal hydrofoils had a mild Re_c dependence, attributed to the blunt trailing edge, and Type II CFRP hydrofoils had a stronger incidence and Re_c dependence. The natural frequency under stall conditions of all but one of the CFRP hydrofoils was in agreement with added mass and finite element analysis estimates. The disagreement was observed in the CFRP hydrofoil with layers aligned at 30° and is attributed to the complex behavior of the carbon layers and to the coupled bend–twist deformation experienced under hydrodynamic loading of the hydrofoil.

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1. Introduction

Hydroelastic tailoring of marine propellers and hydrofoils using composite materials has the potential to improve the hydrodynamic performance of naval ships. In the present context, hydroelastic tailoring is defined as the intentional use of structural and material properties to improve hydrodynamic performance in a broad sense including both static and

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dynamic behavior. Hydrofoils and propellers may be designed to deform under increasing quasi-steady or dynamically varying applied loads to give improved hydrodynamic performance compared with rigid or relatively stiff hydrofoils or propellers. It is envisaged that propellers maybe designed with improved propulsive efficiency as well as reduced unsteady or harmonic force components, vibration and noise emissions. For naval ships, where it is most desirable to reduce vibration and harmonic excitation due to spatial and temporal variations of the inflow to propellers and control surfaces, composites may offer significant advantages over traditional materials.

Traditionally marine propulsion and control equipment have been manufactured in metal particularly using nickel–aluminum bronze or stainless steel. These conventional materials offer the advantages of being homogeneous and isotropic for the purposes of modeling their structural behavior. There is also extensive experience of their design, manufacture and use in the challenging marine environment for both civilian and naval applications (Kerwin, 1986). Composite materials are continuously being developed and are now used extensively in aeronautical applications with limited use to date for ship hull and superstructure applications (Mouritz et al., 2001). Their use for propellers and control surfaces has to date been more limited due to the technical difficulties of designing such devices and the uncertainties over their serviceability and reliability. Nevertheless, composites do offer advantages over traditional materials including reduced weight, corrosion resistance and the potential for hydrodynamic performance improvement through hydroelastic tailoring (Mouritz et al., 2001; Chen et al., 2006; Motley et al., 2009; Miller et al., 2010).

In recent years there has been increased activity focused in computational studies of the fluid–structure interaction (FSI) of deforming hydrofoils and propellers. With increasing sophistication of computational modeling of the turbulent flow about the after-bodies of ships and submarines, the complex nature of the flow in which control surfaces and propellers must operate is being revealed (Fureby, 2007; Andersen et al., 2009; Alin et al., 2010). Several recent studies have presented coupled FSI algorithms and models to study the behavior of composite hydrofoils and propellers in cavitating and sub-cavitating conditions (Young, 2008; Ducoin et al., 2009; Liu and Young, 2009, 2010; Munch et al., 2010; Young et al., 2010; Chae et al., 2013; Akcabay and Young, 2014). Whilst significant progress has been made in this area, comprehensive and detailed experimental data sets are required for further development and validation of computational models.

One of the earliest experiments using flexible hydrofoils was that of Gowing et al. (1998), who reported that the tip deflections helped to delay cavitation inception due to reduced tip loading, while the overall lift and drag coefficients remained unchanged. Ducoin et al. (2009) measured the tip deflection of a hydrofoil made of a plastic material (POM polyacetate) and the results were used to validate a numerical model. The experiments were conducted in a cavitation tunnel with a 192 mm square test section and a hydrofoil with a span of 191 mm. The chord-based Reynolds number ranged from 0.75×10^6 to 1.5×10^6 and incidences from 0° to 6°. The computational results showed that the lift forces exerted on the hydrofoil are strongly coupled with structural deformations. More recently Ducoin et al. (2012b) and Ducoin and Young (2013), using the same foil and experimental setup as previous, showed that for non-cavitating flows the structural response of the hydrofoil depends on the hydrodynamic loading, and that in a transient regime (i.e. transient pitching motion) the tip displacement depends strongly on the pitching velocity.

Given the elastic properties of composite hydrofoils and the documented bending and twisting they experience (Liu and Young, 2009, 2010), it is expected that other forces and moments, particularly the drag force and pitching moment will be significant for validating numerical models and designing flexible hydrofoils and propellers.

To gain basic insight into the FSI problem, particularly as it relates to propellers and control surfaces, experiments of a simple unsteady flow about a flexible three-dimensional hydrofoil have been conceived. The use of a hydrofoil significantly reduces the complexity of models and the experimental setup required compared with that for propellers. An unsteady flow that provides a simplified analogy to spatial and temporal non-uniformity of inflow to propellers or control surfaces is impulsive or periodic variation of incidence. This method eliminates the difficulty of generating unsteadiness in the upstream flow. This arrangement is also compatible with setting up relatively simple computational models. However, it is necessary to first have a baseline knowledge on the hydroelastic behavior of the hydrofoils under steady loading to be able to make meaningful comparisons.

This paper presents the results of an experimental study of the steady hydroelastic behavior of a series of hydrofoil models of nominally identical geometry but of differing materials. In the present context, steady FSI refers to the condition where the structural deformations and flow field are time invariant, as opposed to dynamic FSI where the structural response and flow field vary in time due to either temporal incidence variation or large scale flow fluctuations. This is consistent with the distinctions made by Hodges and Pierce (2011) between static aeroelasticity as an interaction between aerodynamics and elasticity, structural dynamics as an interaction between elasticity and dynamics and dynamic aeroelasticity as an interaction between all three phenomena. The present work is principally concerned with steady deformations in nominally steady flow. Tests were however performed beyond stall where unsteady loads induced unsteady deformations and basic frequency analysis of the hydrofoils were made for completeness.

Models were manufactured in aluminum (AL), stainless steel (SS) and carbon fiber reinforced plastic (CFRP) in two different layups. The use of metal, in addition to CFRP, with predictable material and structural properties provided results that may be compared with the CFRP behavior in both the present experiments and future computational modeling. Measurements were made at several mean chord-based Reynolds numbers up to 10^6 and incidences beyond stall. The results include measurements of lift and drag forces, pitching moments, tip transverse and twisting deformations, and frequency response. This work is limited to steady, non-cavitating flow and is intended as a baseline study for future experiments under oscillating incidence and/or cavitating conditions.

2. Experimental setup and techniques

2.1. Experimental facility

Experiments were carried out in the Cavitation Research Laboratory (CRL) variable pressure water tunnel at the University of Tasmania. The tunnel test section is 0.6 m square by 2.6 m long in which the operating velocity and pressure ranges are 2–12 m/s and 4–400 kPa absolute respectively. The tunnel volume is 365 m^3, which is filled with demineralized water (conductivity of order 1 μS/cm). The tunnel has ancillary systems for rapid degassing and for continuous injection and removal of nuclei and large volumes of incondensable gas. The test section velocity is measured from one of the two (low and high range) Siemens Sitrans P differential pressure transducer models 7MF4433-1DA02-2AB1-Z and 7MF4433-1FA02-2AB1-Z (measuring the calibrated contraction differential pressure) with estimated precisions of 0.007 m/s and 0.018 m/s respectively. The velocity and pressure in the test section are controlled to maintain a constant Reynolds number and Cavitation number. The test section velocity is spatially uniform to within $\pm 0.5\%$, has temporal variations of less than 0.2%, and the free stream turbulent intensity is about 0.5%. Detailed descriptions of the facility are given in Brandner et al. (2007) and Doolan et al. (2013).

2.2. Hydrofoil details

Hydrofoil geometry and physical/mechanical properties have been selected based on the requirements for modeling of static and dynamic conditions typical of those experienced by propellers and hydrofoils operating in ship or submarine wakes. All hydrofoil models are of identical planform but two slightly different section profiles were chosen, a standard NACA0009 and a modified NACA0009. The latter has thicker trailing edge for improved manufacture of the CFRP models. The hydrodynamic characteristics of the NACA0009 profile are comprehensively described in the existing literature. A NACA 4 digit profile was chosen for the hydrofoils as being relatively independent of Reynolds number (Jacobs and Sherman, 1937) particularly for the small incidence range of interest for the future dynamic FSI experiments involving incidence oscillation.

The model sizes have been chosen as a trade-off between Reynolds number and instrumentation limits, and are such that confinement effects should not be significant (Garner et al., 1966; Hackett et al., 2000). The latter are small, particularly with respect to the incidence ranges ($\pm 5^\circ$) of interest for the future oscillating experiments. Computational models will also be developed with the same flow domain as for the experiments. On this basis no confinement corrections have been applied to the present results.

The hydrofoil models of the standard and modified section profiles are referred to as Type I and Type II respectively. The geometry of both hydrofoil types is an upright or unswept trapezoidal planform of 0.3 m span, 0.12 m base chord, and 0.06 m tip chord (aspect ratio 3.33, typical of marine propellers), as shown in Figs. 1 and 2. An unswept geometry was deliberately chosen to principally consider bending deformations only. The scale of the models was chosen to be compatible with the water tunnel test section with the span being half the cross-section dimension and the chord sufficient to obtain chord-based Reynolds number (Re_c) values of at least 1×10^6. $Re_c = U_\infty c/\nu$, where U_∞ is the free stream velocity, c is the hydrofoil mean chord, and ν is the kinematic viscosity of the flowing liquid. Two Type I hydrofoils were manufactured, one in SS (316L) and another in AL (6061 T6). These will be referred to as "Type I-SS" and "Type I-AL", respectively. Both models were machined from solid billets with an integral mounting flange to minimize mass (Fig. 1). The AL model was anodized to a thickness of about 5 μm. The maximum mean side or normal load permissible was set at 1 kN, which gives a maximum stress of about 0.5 of the yield strength for stainless steel, providing an appropriate margin against structural damage inside the water tunnel.

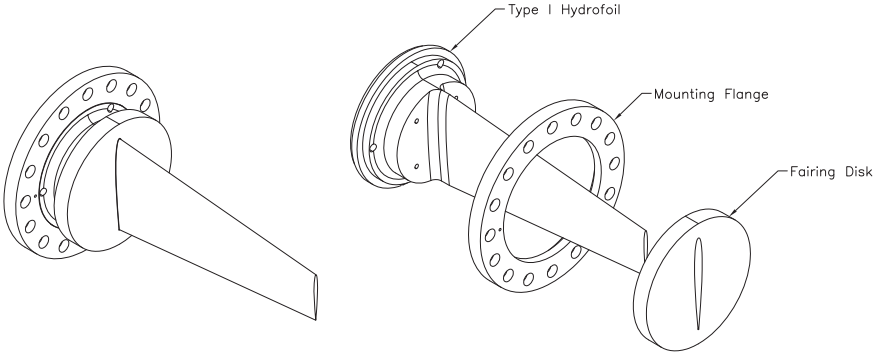


Fig. 1. General arrangement of the Type I hydrofoil model showing mounting flange and fairing disk where the model penetrates the tunnel ceiling.

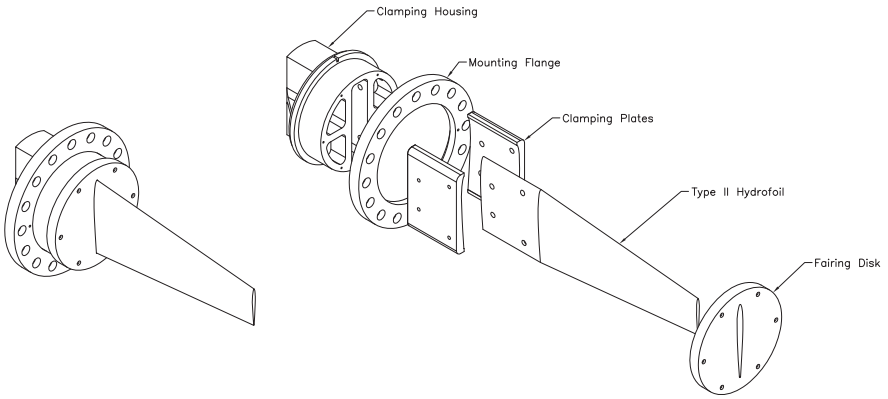


Fig. 2. General arrangement of the Type II hydrofoil model showing clamping housing allowing continuity of the reinforcing fibers for the CFRP models.

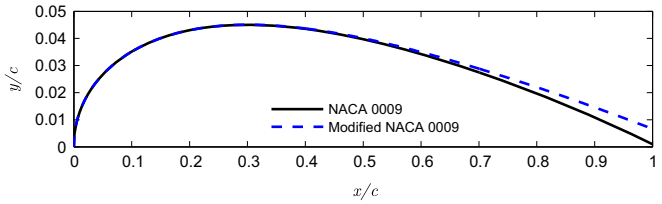


Fig. 3. Half cross-section of the Type I (standard NACA 0009) and Type II (modified NACA 0009) hydrofoil geometry depicting the thin and thick trailing edges, respectively. This modification has the effect of increasing the thickness gradually from about mid-chord to a maximum at the trailing edge. Increasing the thickness there from 0.2% to 1.3%*c*.

For the Type II hydrofoils the trailing edge was made thicker by adjusting the coefficient of the last term in the NACA 4 digit equation as noted below:

standard equation

$$y = 5t(0.2969x^{0.5} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4), \tag{1}$$

modified equation

$$y = 5t(0.2969x^{0.5} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.08890x^4), \tag{2}$$

where t is the thickness to chord ratio (0.09 in this case). This modification has the effect of increasing the thickness gradually from about mid-chord to a maximum at the trailing edge, increasing the thickness there from 0.2% to 1.3%*c*, as shown in Fig. 3.

Four Type II hydrofoils were manufactured: Type II-SS, Type II-AL, Type II-CFRP00, and Type II-CFRP30. The accompanying number in the CFRP model designation indicates the alignment of the carbon unidirectional fibers. Details of the composite construction are provided in the following section. In addition to the modified section, the Type II hydrofoil models were designed with an extended base section into which the reinforcing fibers are run continuously to provide sufficient clamping length to achieve the cantilevered structural boundary condition. As shown in Fig. 2, profiled plates are used to clamp the model within a housing incorporating the mounting flange. Given the similarity of the Type I and Type II sections the maximum mean side or normal load for the latter was also set at 1 kN.

Both metal and CFRP hydrofoils were manufactured to a ± 0.1 mm surface tolerance and a 0.8 μm surface finish. No roughness was added for the purposes of turbulence stimulation to mitigate the effects of changes in Reynolds number, as significant laminar flow effects are not expected for the sections used and the Reynolds number range tested (Jacobs and Sherman, 1937).

2.3. Composite hydrofoil construction

The two CFRP models were manufactured as carbon/glass-epoxy hybrid structures using a resin transfer moulding process. The matrix resin was a very low viscosity, long pot life, two-part epoxy system (Kinetix R118/H103) intended for resin transfer molding (RTM). The structural component of the hydrofoils was made from layers of T700 unidirectional carbon fiber and non-crimp biaxial E-glass fabrics. Two additional fabrics were also used in the hydrofoil construction. A fine basket weave E-glass fabric was placed as the outermost layer of the hydrofoil to aid the surface quality. A sandwich glass mat consisting of two continuous filament random pattern E-glass layers with a polyolefin scaffold core (180 g/m^2) was placed in the center of the hydrofoil. The sandwich mat has an open structure which improves resin flow during the RTM

Table 1
Description of the fabric layers used to manufacture the two composite hydrofoils.

Layer	Carbon	Glass	Glass	Glass
Material	12k T-700	E-glass	E-glass	E-glass/Polyolefin
Type	Unidirectional (0°)	Biaxial (0°/90°)	Basket (0°/90°)	Continuous filament skins/polyolefin scaffold core
Weight (g/m^2)	300	600	130	780
Thickness (mm)	0.25	0.6	0.15	≈2.0

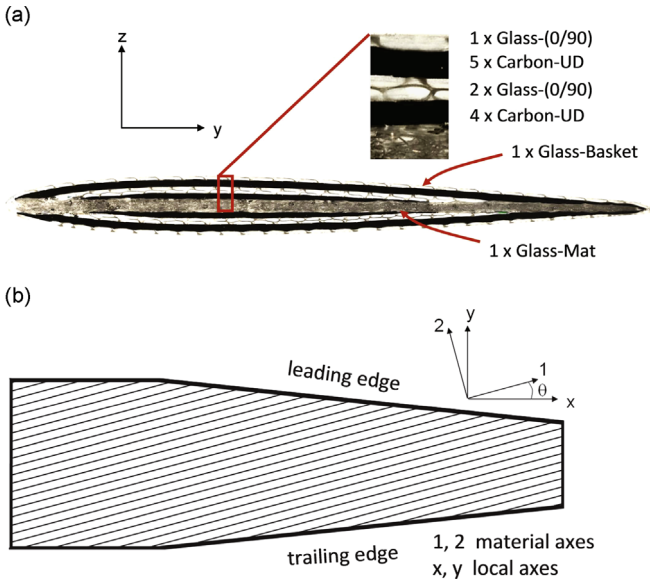


Fig. 4. (a) Root cross-section showing the layered structure of the CFRP hydrofoils. (b) Schematic plan view showing the material, local co-ordinate systems used, and alignment of the carbon unidirectional layers.

process. The properties of these fabrics are summarized in Table 1. Fig. 4 shows a typical cross-section of the hydrofoil root as well as the local and material coordinate system. The stacking sequence of the structural layers consisted of alternating blocks of glass (0°/90°) and carbon unidirectional layers. The Type II-CFRP00 hydrofoil had the carbon unidirectional layers aligned along the spanwise axis of the foil, $\theta=0°$, and had the stacking sequence $[(0°/90°)_g, (0°)_{5c}, (0°/90°)_{2g}, (0°)_{4c}]_S$, where the subscripts g and c refer to the number of glass (0°/90°) and carbon unidirectional layers respectively, and the subscript S refers to the structural layers being symmetrical around the hydrofoil mid-plane. The Type II-CFRP30 hydrofoil had the carbon unidirectional layers off-set toward the leading edge ($\theta=30°$) and had the stacking sequence $[(0°/90°)_g, (30°)_{5c}, (0°/90°)_{2g}, (30°)_{4c}]_S$. The relevant coordinate systems are shown in Fig. 4. In both cases the hydrofoil profile and spanwise taper were accommodated by dropping plies internally to guarantee that the longest layers were on the outside of the hydrofoils.

2.4. Structural properties of the hydrofoils

The hydrofoil materials were chosen as suitable in terms of desired properties and being practical to manufacture. Table 2 summarizes the material and structural properties. The ratio of hydrofoil to liquid density or mass ratio, ρ_H/ρ_L, differs by about a factor of 5–2 between the three hydrofoil materials.

Impact testing was carried out to estimate the natural frequencies of the hydrofoils and the force balance used in the experiments. The natural frequency of the hydrofoils was measured before mounting them in the water tunnel. The composite hydrofoils were also tested after the hydrodynamic loading to verify that no gross internal (fiber) failure had taken place. Each of the hydrofoils was fastened to a rigid mount of large mass and suspended from long compliant (polyamide) lifting strops to guarantee that the rigid body frequency was much lower than the natural frequencies of the hydrofoils. The hydrofoils were fastened using the same mounting arrangement shown in Figs. 1 and 2, for Type I and II hydrofoils respectively. A DeltaTron accelerometer (Brüel & Kjær, type 4507 B 004), attached to the hydrofoil mid-chord at a spanwise distance of 50 mm from the tip, was used to measure the frequency response. The hydrofoils were impacted at approximately the mid-chord/mid-span point with an instrumented modal hammer (Meggitt-Endevco, model 2302-50) fitted with a soft tip. The input pulse and hydrofoil response data were acquired using a NI-LabVIEW (National Instruments) software interface and NI hardware. The signals were sampled at 2^{14} Hz and the sample period was 2 s. The sampling frequency was high enough to resolve the input pulse and the sampling period long enough to capture the exponential

Table 2
Summary of material and structural properties of the hydrofoils.

	Hydrofoil					
	SS		AL		CFRP00	CFRP30
Type	I	II	I	II	II	II
K (N/mm)	61.7	60.2	23	22.1	20.0	8.2
E (GPa)	193	193	71	71	65	26
I (mm^4)	5956	6148	5956	6350	6148	
J ($mm^4 \times 10^3$)	860.4	854.5	860.4	854.7	854.5	
ρ_H (kg/m^3)	7900		2700		1600	
f_n (Hz) – air	100	96	100	96	112	72
f_n (Hz) – water	62	61	42	41	41	26
ρ_H/ρ_L	7.9	7.9	2.7	2.7	1.6	1.6

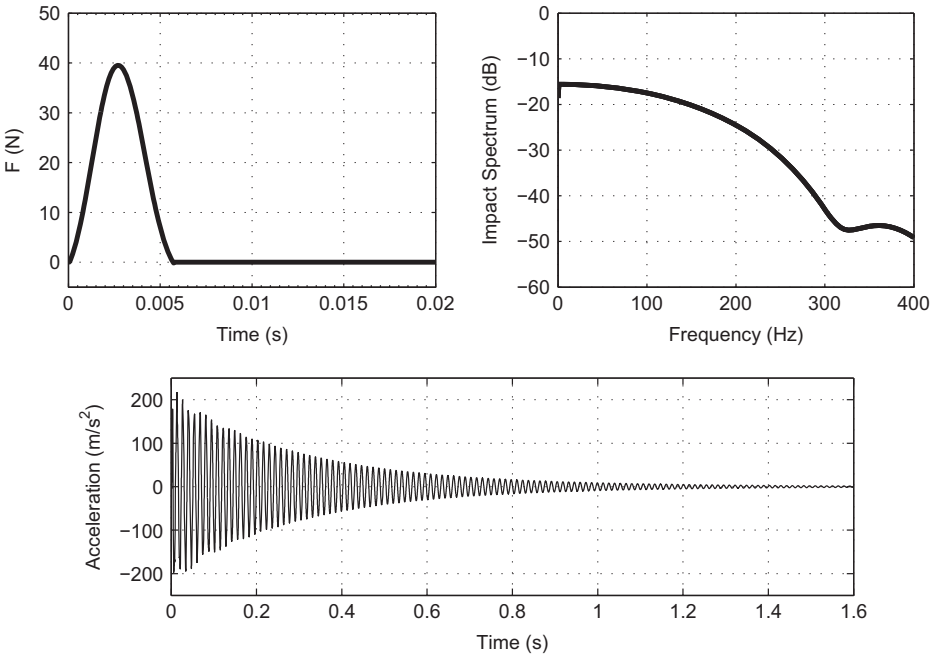


Fig. 5. Impact input pulse (*left*) and spectrum (*right*) and frequency response (*bottom*) of the Type II-CFRP30 hydrofoil.

decay to within less than 1% of the initial response. Fig. 5 shows the results from the impact test carried out for the Type II-CFRP30 hydrofoil. The input pulse data was cropped to capture the input signal only and a rectangular window was applied as the time series was sufficiently long to meet the six-time constant data capture criteria (Trethewey and Cafeo, 1992). The spectrum of the input pulse shows that signal amplitude did not decrease more than 5 dB at the first mode natural frequency of the hydrofoils (discussed below), hence making the frequency response from the input pulse negligible. Similar results and behavior were observed from the impact testing for all the model hydrofoils. Fig. 6 shows the magnitude of frequency response function (Rao, 2011) of the composite hydrofoils, $|H(f)|$, where H describes the relationship between the autospectral density function of the input pulse and response of the hydrofoils.

The natural frequency (f_n) in air for all the hydrofoils is listed in Table 2. SS and AL have similar ratios of elastic modulus (E) to density (ρ_H) such that for the same geometry they have virtually identical natural frequencies in air. The hydrofoil profiles were chosen as a compromise between the level of flexibility and response and maintaining a reasonable margin against hydroelastic instability phenomena. The chosen section thickness gives about 100 Hz first mode natural frequency in air for both hydrofoil geometries, and 62 and 42 Hz in water for the SS and AL models, respectively (from added mass estimate, Blevins, 1979). The small differences in natural frequencies between the metal hydrofoils of each type may be attributed to the differences in section profile and potentially the differences in mounting. The first mode natural frequency of the CFRP00 and CFRP30 hydrofoils was 112 and 72 Hz in air, respectively.

The first four bending modes in air and corresponding shape of the CFRP hydrofoils were estimated using the commercial Finite Element Analysis (FEA) software Strand7 and are presented in Fig. 7. The fourth mode for the CFRP00 hydrofoil and the first, second and fourth modes for the CFRP30 hydrofoil correspond to mixed shapes, which refer to a combined bending

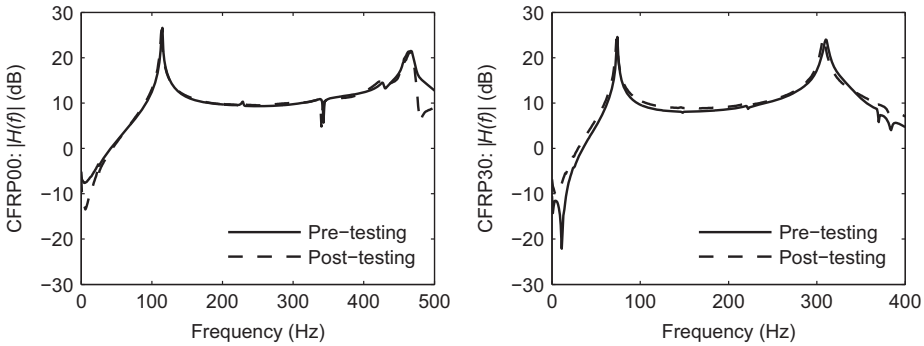


Fig. 6. Frequency response function of the Type II-CFRP00 (*left*) and Type II-CFRP30 (*right*) hydrofoils. Frequency response was measured pre- and post-hydrodynamic loading to verify that no internal failure had occurred.

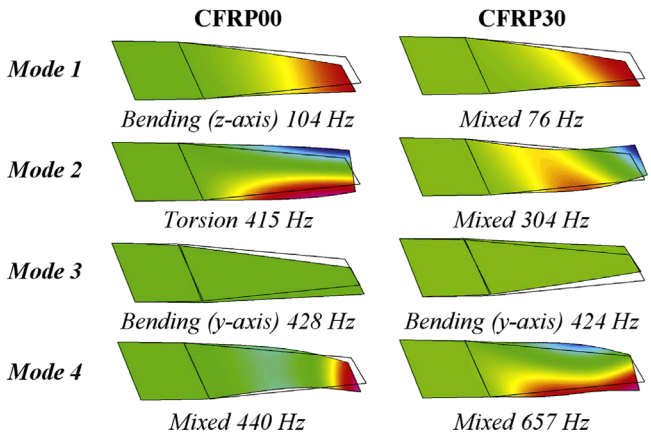


Fig. 7. Modal shape and frequency of the first four bending modes in air for the two composite hydrofoils derived from the FEA model. A mixed shape refers to a combined bending and torsion deformation. Mode 3 (bending in the y direction) may also be referred to as lead-lag mode. The coordinate system is the same as shown in Fig. 4.

and torsion deformation. The FEA model consisted of 7501 quadrilateral (Quad4) shell elements centered on the hydrofoils mid-plane. The laminate feature in Strand7 was used to assign material properties to the shell elements. The profile and taper of the hydrofoils were accounted for by dropping the innermost plies first according to the local thickness of each element. The analysis was carried out using an infinitely rigid boundary condition at the root of the hydrofoil.

In order to assess the structural integrity of the composite hydrofoils an additional CFRP00 model was manufactured and tested under cantilever loading until failure was detected. Failure occurred at a load of 2.49 kN by delamination at the thickest section near the root. The failure site correlated to a region of high inter-laminar stress observed at the same position by FEA. Further details about these experiments can be found in Ibrahim et al. (2014).

The bending stiffness, $K = F/\delta$, was determined experimentally using an end loaded cantilever arrangement, where F is a point load applied at the tip of the hydrofoil and δ is the average tip displacement. The displacement was measured with two LVDT transducers (Solartron, G-Series). These were located 55 mm apart equi-spaced about the hydrofoil mid-chord 3 mm from the tip. The load was applied at a quarter-chord from the leading edge and in line with the transducers. The quarter-chord loading location was chosen because it coincides with the estimated location of the center of pressure of the hydrofoils under hydrodynamic loading. The leading and trailing edge tip deflections were measured for zero and three ascending point loads. In all cases the deflection presented a linear relationship. Therefore, it was possible to estimate K from the linear fit of the F–δ relationship, where δ is the mid-chord tip deflection. The bending stiffness of each hydrofoil is presented in Table 2.

The bending stiffness results were used to estimate an elastic modulus E for each material (Table 2), by using the known modulus for SS ($E = 193$ GPa) as a reference. For isotropic materials such as AL and SS, the dimensionless deflections $\delta' = \delta EI/(F\ell^3)$, where ℓ is the distance to the point load, I is the base section second moment of area, should be the same for each hydrofoil type under the same loading. Using this equation, the derived elastic modulus for AL was within the manufacturer supplied range. Accordingly, equivalent moduli were derived for the CFRP models, as these do not have isotropic properties. The I values and the torsional constants, J, for each of the hydrofoils (based on dimensions from metrology for each finished model calculated using computer aided drafting software) are listed in Table 2. Using these modulus values, all the measured deflections can be presented in a dimensionless form for comparison, as presented below.

From Table 2 and Fig. 7 it is seen that the experimental and numerical results for the natural frequency of the composite hydrofoils are in good agreement. Note also that the second modes were observed during the impact test, as shown in Fig. 6. However, the second mode for the CFRP00 hydrofoil (467 Hz) does not match the FEA results. This can be attributed to the accuracy of the impact test at that frequency range. The magnitude of frequency content of the input pulse decreases considerably after 400 Hz and the results are no longer reliable. In comparison, the second mode of CFRP30 hydrofoil (310 Hz) is in good agreement, given that it occurs below 400 Hz.

From the FEA results in Fig. 7 it can be observed that the two composite hydrofoils have different mode shapes. The first three modes in the CFRP00 hydrofoil are pure bending or twisting, while the fourth mode is a combined one. On the other hand, the CFRP30 hydrofoil undergoes a combined bending and twisting deformation at the first, second and fourth modes, and pure bending for the second, due to the reduction in the bending stiffness compared with the CFRP00. These characteristics cause a significant difference in the behavior of the two hydrofoils under hydrodynamic loading conditions, as shown in the Results section below.

2.5. Techniques and conditions

The models were mounted on a six-component force balance extending vertically into the flow through a 0.16 m diameter penetration on the tunnel ceiling. The 0.16 m diameter penetration was made fair (to 50 μm) using a disk mounted on the hydrofoil (see Figs. 1 and 2) and therefore on the measurement side of the force balance, with a nominal 0.5 mm radial clearance to avoid interference with the force measurement. Of the total load vector measured using the force balance, mean and unsteady components of lift, drag and pitching moment are presented. Spanwise forces and roll/yaw moments although measured are not presented as they are of secondary interest to the present work and they also include forces/moments due to the pressure field acting on the fairing disk. Data were sampled at 1 kHz for durations from $T_s = 10$ to 30 s for the highest to lowest Re_c values, respectively. The sampling time was adjusted so that $T_s U_\infty / c$ was similar in all the test cases. That is, data was acquired for a similar number of chord lengths traveled by the flow in each case.

Type I hydrofoils were tested at two streamwise locations, 0.7 and 1.3 m from the test section entrance, to assess the effect of varying ceiling boundary layer thickness. Results showed the effect of the boundary layer thickness on the lift to be less than 1%. The boundary layer thickness (for $0.99U_\infty$) at the 0.7 and 1.3 m positions is about 19 and 26 mm respectively. Type II hydrofoils were tested at 0.7 m only. The results for Type I hydrofoils at 1.3 m from the test section entrance are omitted here and can be found in Brandner and Pearce (2012).

The force balance is calibrated by a least squares fit between a basis vector loading cycle and the six outputs giving a 6×6 matrix from which estimated precision on all components is less than 0.5%. Forces were measured at mean chord-based Re_c values (mean chord $c = 0.09$ m) of 0.2, 0.4, 0.6, 0.8 and 1.0×10^6, at incidence (α) values beyond stall or up to the maximum load of 1 kN. For the lowest three Re_c values the incidence was incremented beyond stall to 15° without exceeding the maximum load of 1 kN. Whereas, for $\mathrm{Re}_c = 0.8$ and 1.0×10^6 incidences were limited to 9° and 6° respectively to avoid exceeding the limit. Hydrofoil incidence is adjusted using the force balance automated indexing system, here incremented in 0.5° steps. The absolute position of the indexing system is less than 0.1° and the incremental precision is less than 0.001°. For hysteresis tests a 0.05° incidence correction is applied for backlash in the indexing mechanism. The water tunnel was pressurized up to 200 kPa for all tests to prevent cavitation occurrence. The blockage ratio, defined as the quotient of the planform area of the hydrofoils and the tunnel cross-sectional area, is 0.075. For the range of testing conditions, the blockage effect on the results is expected to be negligible (Mueller and Batill, 1982; Hackett et al., 2000).

Tip deflections were measured using the image registration algorithm described by Guizar-Sicairos et al. (2008). Still photographs of the hydrofoil tip faces before and after loading were interrogated with the algorithm. To improve the deflection measurements, two contrasting targets were placed near the leading and trailing edges of the tip faces (Fig. 16). Photographs were taken using a Canon EOS 50D 35 mm digital SLR camera with a Canon EF 24–70 mm lens using natural lighting (image resolution 4752×3168 pixels). The images were calibrated using the known tip-chord length of the hydrofoil. Deflections were measured for all Re_c mentioned above up to 10° incidence in 2° increments. The uncertainty of the measurement (bias and random error) was estimated, based on the Type II-CFRP30 results, with a worst case uncertainty of 13% at $\alpha = 2°$ and $\mathrm{Re}_c = 0.2 \times 10^6$ and best case of 2% at $\alpha = 6°$ and $\mathrm{Re}_c = 1.0 \times 10^6$. The average uncertainty of the deflection measurements was estimated to be 3.2%.

3. Results

The results are presented in two sections, forces and deformations. The first describes measured mean and unsteady forces, including hysteresis loops, root-mean-square (RMS) forces and force fluctuation power spectrum. Forces and moments are presented in either flow fixed or body fixed coordinate systems. The coordinate system origin in each case is located at the hydrofoil mid-chord, aligned with the flow direction or chord line for the flow and body fixed coordinate systems, respectively. The lift and drag forces and pitching moments are presented as dimensionless coefficients, $C_L = 2L/(\rho_L U_\infty^2 sc)$, $C_D = 2D/(\rho_L U_\infty^2 sc)$ and $C_M = 2M/(\rho_L U_\infty^2 sc^2)$, respectively. Likewise, the normal and axial forces are defined as a coefficient, $C_N = 2N/(\rho_L U_\infty^2 sc)$, $C_A = 2A/(\rho_L U_\infty^2 sc)$. U_∞ is the freestream velocity, s the hydrofoil span and c the mean chord.

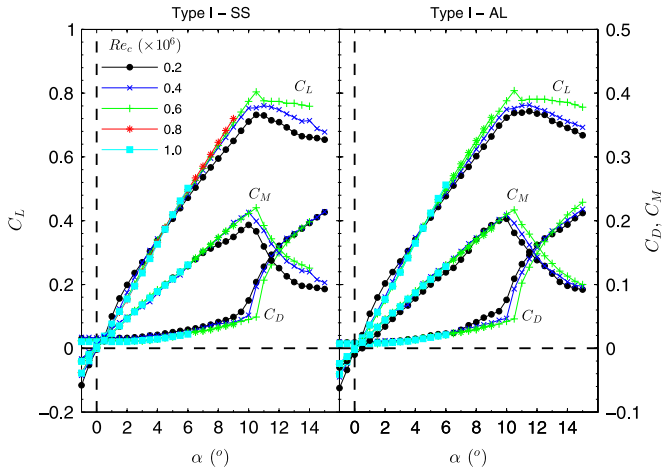


Fig. 8. Lift, drag and pitching moment coefficients with incidence for Type I-SS (*left*) and Type I-AL (*right*) hydrofoil models, for several Re_c values.

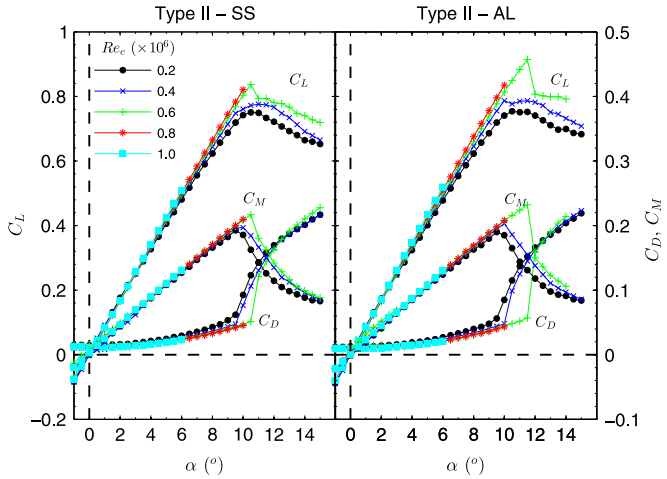


Fig. 9. Lift, drag and pitching moment coefficients with incidence for Type II-SS (*left*) and Type II-AL (*right*) hydrofoil, for several Re_c values.

The second section reports the measured hydrofoil tip deflections, δ, and the observed twisting, γ, of the CFRP hydrofoils. Results from the Type I hydrofoils were reported previously by Brandner and Pearce (2012) and only some results are reproduced here for comparison.

3.1. Forces

Mean coefficients of lift, drag and pitching moment for Type I/II-SS and Type I/II-AL hydrofoils are presented in Figs. 8 and 9. Forces and moments for both types of SS and AL hydrofoil models appear to compare closely for all Re_c and α values. Stall occurs at nominally 10.5° incidence for the three Re_c values tested for both types. Pre-stall lift forces show a dependence on Re_c for low values, due presumably to laminar flow effects, but become independent for $Re_c = 0.8 \times 10^6$ and greater for the incidence range tested. Similar behavior is evident in the drag and pitching moment coefficients. This shows that deflections have little effect on the hydrodynamic forces for the metal hydrofoils. Despite the Reynolds number independence for pre-stall incidences reported by Jacobs and Sherman (1937), the results for the type I hydrofoils show effects for the lowest Reynolds number ($Re_c = 0.2 \times 10^6$) that suggest complex laminar flow effects. Ohtake et al. (2007) and Selig et al. (1995) show similar effects for two-dimensional NACA 4-digit sections for Re_c up to 10^5.

The behavior of the CFRP foils differ significantly from that of the metal hydrofoils. Fig. 10 shows the mean coefficients for the Type II-CFRP00 and CFRP30 hydrofoils. It is observed that in both hydrofoils, the force coefficients have differing Re_c dependence. The CFRP00 hydrofoil experiences an increase in lift with increasing Re_c, while the drag and pitching moment show little or no pre-stall Re_c dependence. On the other hand, the CFRP30 hydrofoil force and moment coefficients decrease with increasing Re_c for the linear part of the lift curve. For incidences approaching and exceeding stall the Re_c dependence reverts to the same behavior as the other foils. The stall behavior for the CFRP00 hydrofoil is similar to the metal hydrofoils, occurring near 10.5°, whilst for the CFRP30 stall is delayed with increasing Re_c.

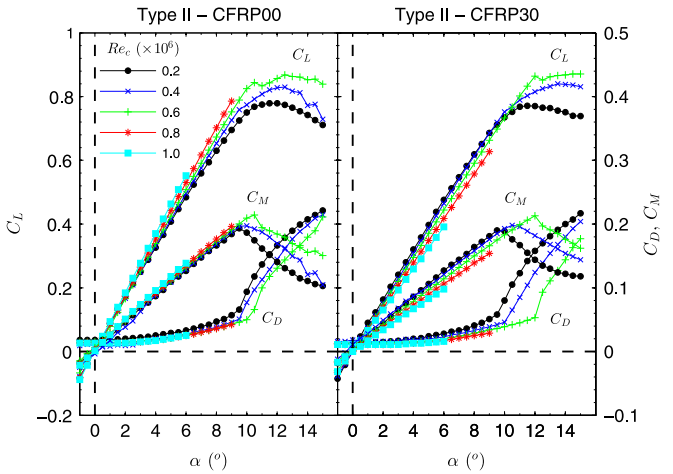


Fig. 10. Lift, drag and pitching moment coefficients with incidence for Type II-CFRP00 (*left*) and Type II-CFRP30 (*right*) hydrofoil models, for several Re_c values.

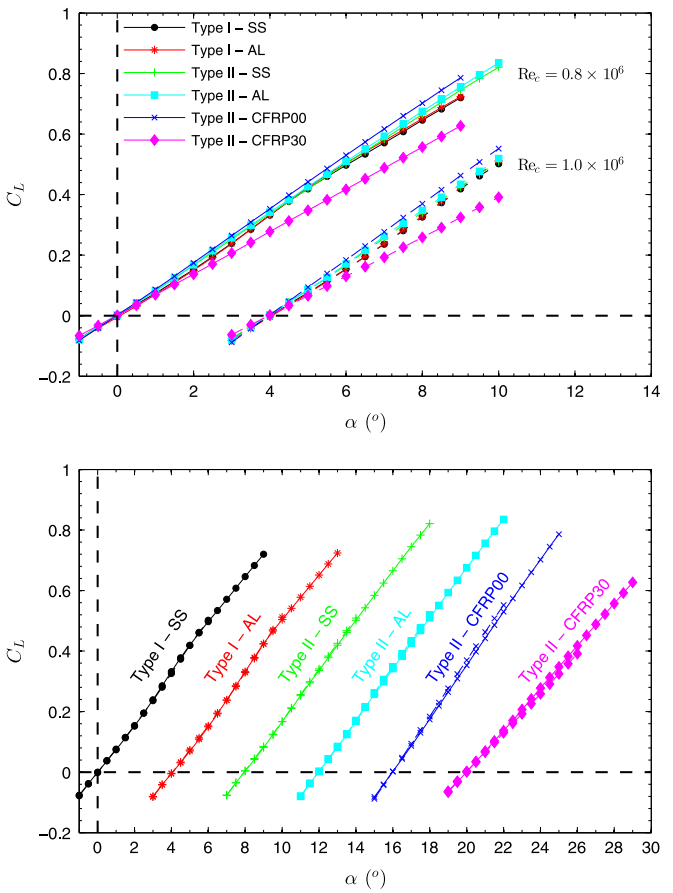


Fig. 11. Comparison of lift forces with incidence at the two highest Reynolds numbers, $Re_c = 0.8 \times 10^6$ (*solid lines*) and $Re_c = 1.0 \times 10^6$ (*dashed lines*). Both panels present the same results: the top shows the differences in lift curves for a fixed Reynolds number and the bottom the effect of Reynolds number for each hydrofoil. The top panel shows how the CFRP00 and CFRP30 hydrofoils have increased and decreased lift slope, respectively, compared with the SS and AL hydrofoils. The bottom panel shows the Re_c independence of SS and AL hydrofoils and the Re_c dependence of CFRP hydrofoils. Plots are staggered by increments of 4°.

The lift coefficient data for the two largest Re_c values are replicated in Fig. 11. These demonstrate the Re_c independence for the SS and AL hydrofoils despite the difference in deflections (discussed in the next section), and the Re_c dependence of the CFRP hydrofoils. A slight difference in lift coefficient between the SS and AL hydrofoils of each type is just evident as the

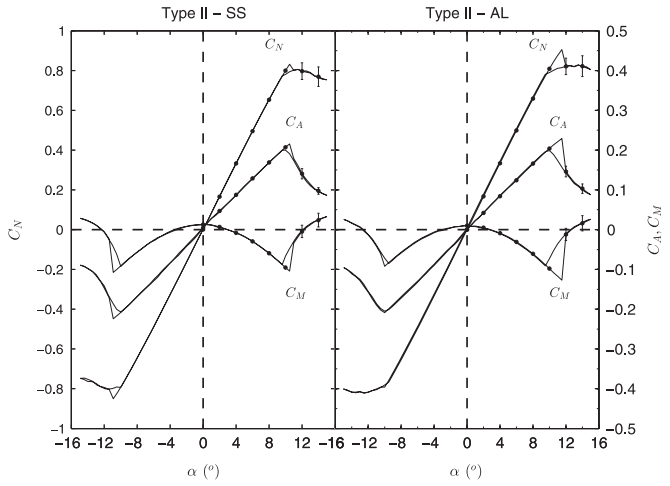


Fig. 12. Foil chordwise normal (C_N) and axial (C_A) force and pitching moment (C_M) measurements taken over a full incidence cycle (α=0° to +15° to −15° to 0°) for Type II-SS and Type II-AL hydrofoils indicating no hysteresis ($Re_c = 0.6 \times 10^6$). Also shown as an error bar is the RMS of the force/moment coefficients at selected values of positive incidence.

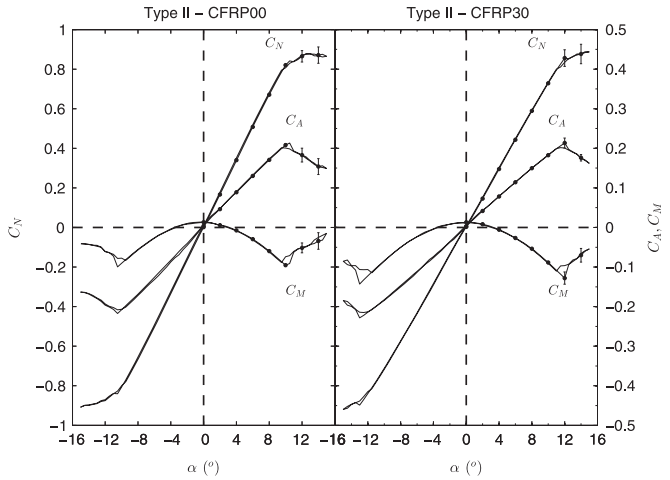


Fig. 13. Foil chordwise normal (C_N) and axial (C_A) force and pitching moment (C_M) measurements taken over a full incidence cycle (α=0° to +15° to −15° to 0°) for Type II-CFRP00 and Type II-CFRP30 hydrofoils indicating no hysteresis ($Re_c = 0.6 \times 10^6$). Also shown as an error bar is the RMS of the force/moment coefficients at selected values of positive incidence.

incidence increases. This indicates that for higher incidences, differing deflections may affect the lift. The differences in behavior between the composite and metal models are due to the type of deformation sustained by each hydrofoil. Also shown in Fig. 11 is the increased and decreased lift slope for the CFRP00 and CFRP30 hydrofoils, respectively. The change in loading conditions for the CFRP hydrofoils is explained by their torsional deformation under hydrodynamic loading. The deformations are presented in the next section and their effect on the hydrodynamic performance of the hydrofoils is discussed below.

Figs. 12 and 13 present force and moment coefficients, in body fixed coordinates, for a complete incidence loop for the Type II hydrofoils at $Re_c = 0.6 \times 10^6$. These show that the presumed laminar flow effects at low Re_c essentially do not result in hysteresis. Results for the Type II hydrofoils are similar to the Type I as reported by Brandner and Pearce (2012). Figs. 12 and 13 also show the RMS of the force and moment coefficients within the positive incidence range. The RMS values show that the unsteady normal forces are less than 1.4% of the mean for all incidences up to stall, after which they suddenly increase up to 6.4% (SS and AL), 4.8% (CFRP00), and 5.6%(CFRP30). It can also be observed that hysteresis effects are only visible at post-stall incidence.

Spectra of the normal force fluctuations are shown in Figs. 14 and 15. At pre-stall incidences there is a level response up to a peak at about 120 Hz, which corresponds to the force balance first mode. For the post-stall spectra the magnitude increases, in agreement with the measured RMS forces. New peaks appear in the spectra at 54, 46, 40, and 53 Hz for the Type II-SS, AL, CFRP00 and CFRP30, respectively. These reflect the excitation of the hydrofoil first mode with the onset of

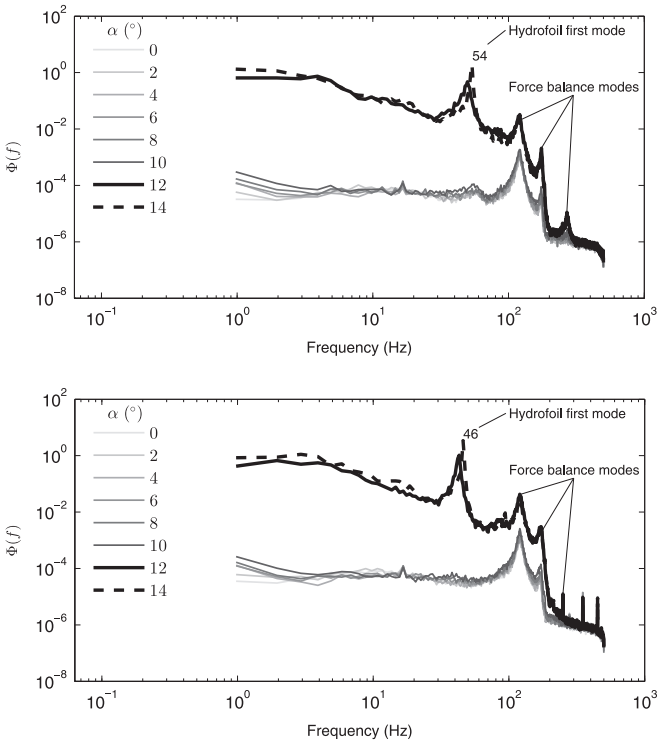


Fig. 14. Power spectral density of normal force fluctuations for Type II-SS (*top*) and Type II-AL (*bottom*) hydrofoils.

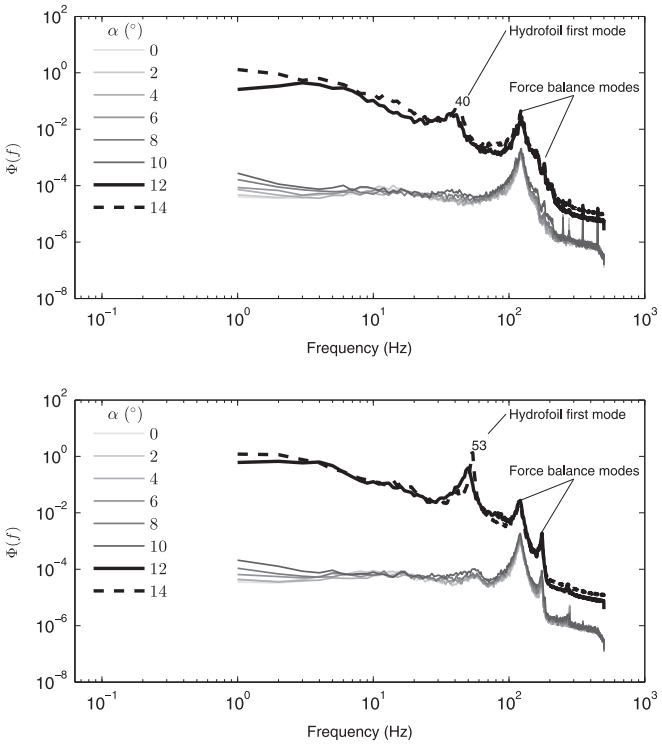


Fig. 15. Power spectral density of normal force fluctuations for Type II-CFRP00 (*top*) and Type II-CFRP30 (*bottom*) hydrofoils.

unsteady flow with stall. It can also be observed that the hydrofoils first mode is apparent in some of the pre-stall spectra. These frequencies agree with those predicted for the hydrofoils, as listed in Table 2, except for the CFRP30 model. The lack of agreement between the estimated first mode for the Type II-CFRP30 and the measured response under hydrodynamic

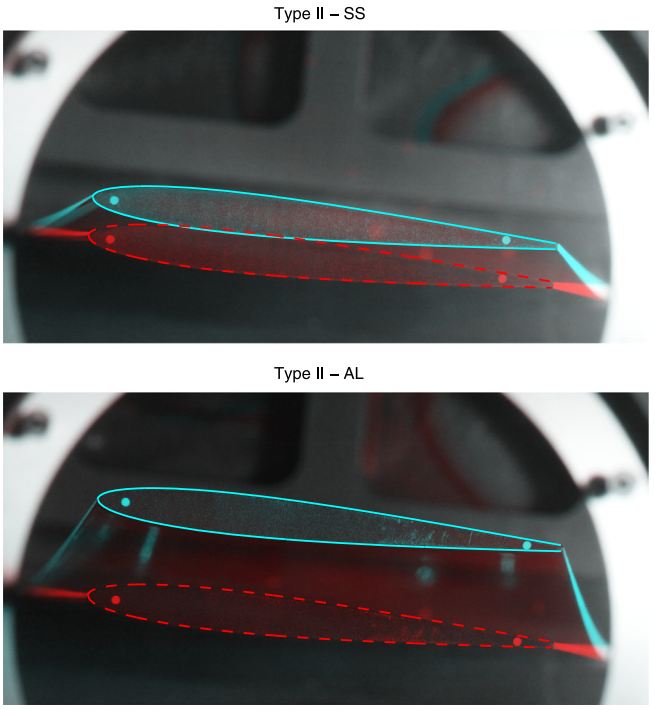


Fig. 16. Overlayed images of the hydrofoil tips with (*solid line*) and without (*dashed line*) flow at 6° incidence and $Re_c = 1.0 \times 10^6$ showing test targets for image cross-correlation to derive tip deflection. The top panel shows the Type II-SS hydrofoil with a deflection of 5 mm. The bottom panel shows the Type II-AL hydrofoil with a deflection of 12.5 mm.

loading suggests a more complex behavior for this hydrofoil. As discussed below this hydrofoil exhibits a significant coupled bend–twist deformation, suggesting that the measured response is not due to bending alone but a coupled mode due possibly to an FSI response (Ducoin et al., 2012a).

Recent studies of free vibrations of cantilever composite plates in air and water, with the same carbon fiber orientation as used in the present study, have shown that the frequency response is 50–70% lower in water than in air, due to large added mass effects (Kramer et al., 2013). This demonstrates that in some cases a simple added mass correction for bending only may not provide an acceptable estimate.

3.2. Deformations

Figs. 16 and 17 show sample image pairs for the largest pre-stall deflections measured for the Type II hydrofoils at $Re_c = 1.0 \times 10^6$ and $\alpha = 6°$. These images were taken using stroboscopic illumination, whereas the images used for cross correlation to measure the deflections were taken using natural lighting. The latter technique resulted in a darker background with mostly the white markers showing, providing a better contrast for the image registration algorithm. As only still images were taken, this technique was limited to pre-stall incidences where there was no observable hydrofoil vibration. It was observed that for the same hydrodynamic conditions, the four hydrofoils have significantly different deformations. The SS hydrofoil, as expected, has the lowest tip deflection, the AL and CFRP00 similar deflections, with CFRP30 having the greatest deflection. The CFRP hydrofoils differ from the metal ones in that they have a mixed bending and twisting deformation. Although not apparent in these images, the CFRP00 is also twisting, as discussed below. The largest deflections and twist angles were observed in the CFRP30 hydrofoil, for all test cases.

The maximum deflections measured at $Re_c = 1.0 \times 10^6$ and 6° incidence were 4.9 mm and 13.4 mm for the Type I SS and AL hydrofoils, 5.0 mm and 12.5 mm for the Type II SS and AL, respectively. For the SS and AL models no twist was resolved within the precision of the method used, suggesting that it is small enough to be negligible. The force measurements tend to confirm this as they show no difference between the two materials despite the much larger deflections (and potentially the twist) of the AL hydrofoil. In contrast, the behavior of the CFRP models is significantly different.

As twist was observed in both CFRP models, the maximum deflection of the leading and trailing edge are reported separately. The Type II-CFRP00 hydrofoil had deflections of 15.4 mm and 14.9 mm, and the Type II-CFRP30 20.8 mm and 22.6 mm, at the leading and trailing edges, respectively. Note that with the CFRP00 model, the leading edge deflection is larger than that of the trailing edge, which indicates that the hydrofoil is experiencing positive twist. In contrast, the CFRP30 model has the opposite behavior, negative twist. The force measurements confirm this as they show an increase in the lift coefficient for the CFRP00 model and a decrease in the CFRP30 model.

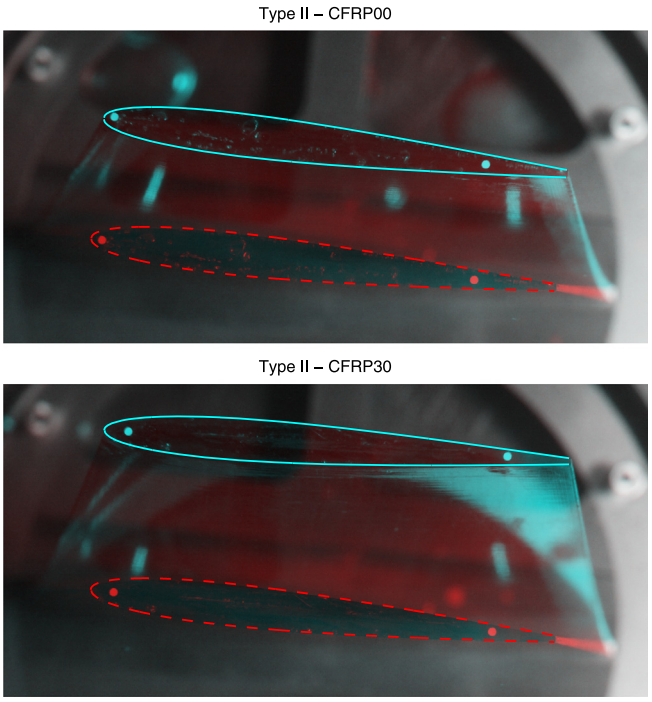


Fig. 17. Overlayed images of the hydrofoil tips with (*solid line*) and without (*dashed line*) flow at 6° incidence and $\mathrm{Re}_c = 1.0 \times 10^6$ showing test targets for image cross-correlation to derive tip deflection. The top panel shows the Type II-CFRP00 hydrofoil with a leading and trailing edge deflections of 15.4 mm and 14.9 mm respectively, and a twist angle of 0.6°. The bottom panel shows the Type II-CFRP30 hydrofoil with a leading and trailing edge deflections of 20.8 mm and 22.6 mm respectively, and a twist angle of −2.2°.

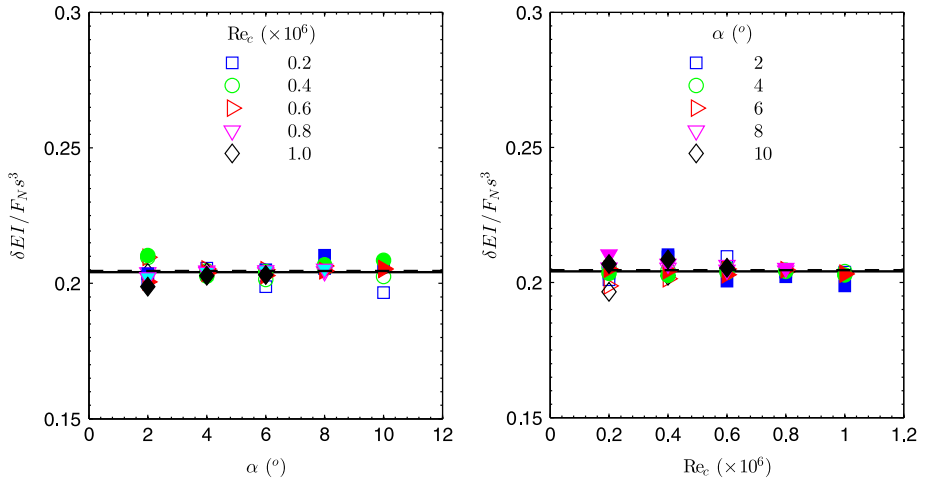


Fig. 18. Comparison of dimensionless mid-chord tip deflections for Type I-SS (*open symbols*) and Type I-AL (*closed symbols*) hydrofoils against incidence (*left*) and Re_c (*right*). SS mean dimensionless deflection, 0.204 (*solid lines*); AL mean dimensionless deflection, 0.202 (*dashed lines*).

Although the magnitude of the tip deflections is a function of the material properties, the dimensionless values, $\delta' = \delta EI/(F_N s^3)$, where F_N is the hydrodynamic force normal to the chord line, are comparable. Figs. 18–21 show the relationship between the dimensionless deflections, incidence angle and Reynolds number for all hydrofoils.

For isotropic materials, the dimensionless deflections should be the same, provided the load distributions are the same. Figs. 18 and 19 for the SS and AL hydrofoils show this result. The Type I results show virtually no dependence on incidence or Re_c allowing for greater error at the lower incidence and Re_c values, as the deflections are smallest. Type II SS/AL hydrofoils show a Re_c dependence, which reduces for $\mathrm{Re}_c > 0.6 \times 10^6$. These values of about 0.2 compare favorably with, for example, 0.125 for a uniformly loaded cantilever of uniform rectangular cross-section. The mean values indicated in Figs. 18 and 19 for the SS and AL hydrofoils show a difference of 1.0% and 3.0%, respectively. The Type II hydrofoils show a Re_c dependence that may be attributed to the blunter trailing edge of this profile compared with the Type I profile (Bourgoyne et al., 2003).

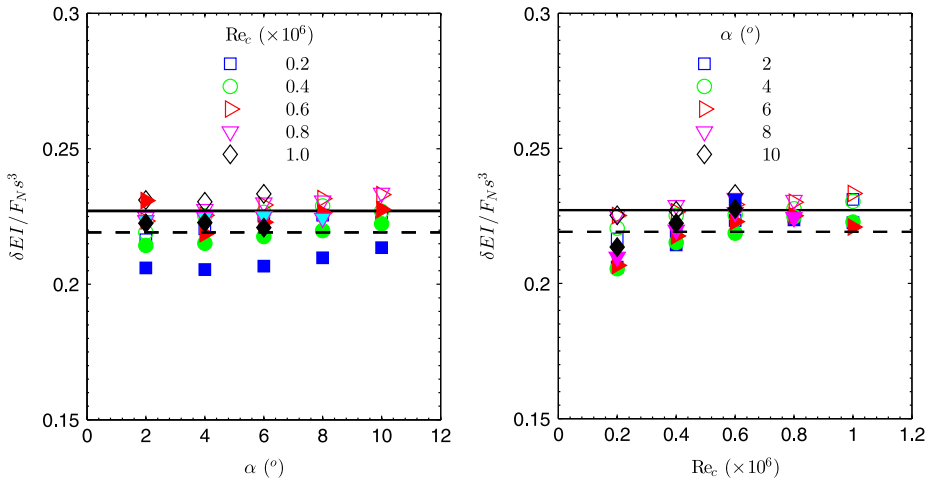


Fig. 19. Comparison of dimensionless mid-chord tip deflections for Type II-SS (*open symbols*) and Type II-AL (*closed symbols*) hydrofoils against incidence (*left*) and Re$_c$ (*right*). SS mean dimensionless deflection, 0.227 (*solid lines*); AL mean dimensionless deflection, 0.219 (*dashed lines*).

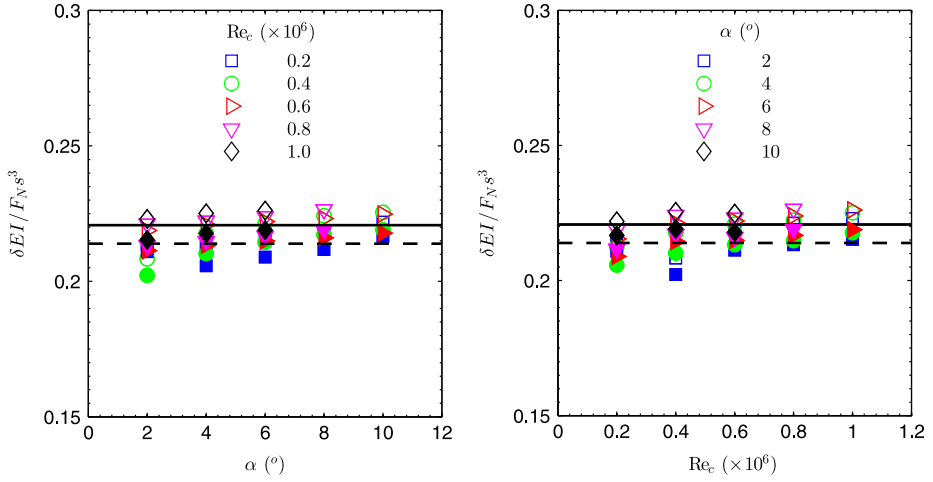


Fig. 20. Comparison of dimensionless leading edge (*open symbols*) and trailing edge (*closed symbols*) tip deflections for the Type II-CFRP00 hydrofoil against incidence (*left*) and Re$_c$ (*right*). Mean leading edge dimensionless deflection, 0.221 (*solid lines*); mean trailing edge dimensionless deflection, 0.214 (*dashed lines*).

The forces for the Type I hydrofoils are Reynolds number independent at low incidences for all but the lowest Re$_c$ value, as shown in Fig. 11. At the higher incidences there is Reynolds number dependence with convergence occurring for the higher values. These however are not reflected in the dimensionless deflection results. For the Type II metal hydrofoils there is a consistent Reynolds number dependence with the lift slope increasing with increasing Re$_c$, converging for the higher values, as shown in Fig. 9. This is also shown in Fig. 19 where the dimensionless deflections converge for Re$_c > 0.6 \times 10^6$.

The composite hydrofoils present a significantly different behavior, particularly the Type II-CFRP30 model. The dimensionless leading and trailing edge deflections for the CFRP00 and CFRP30 are shown in Figs. 20 and 21, respectively. The mid-chord deflections (not shown) for the CFRP00 model have similar behavior to those for the Type I and Type II SS and AL hydrofoils. For the Type II-CFRP30, the deflections show a dependence on incidence, and an Re$_c$ dependence for the higher Re$_c$ values.

Fig. 22 presents the twist measured in the CFRP hydrofoils (top panel) and the dimensionless twist (bottom panel), $\theta' = \theta GJ/Ps^3$, where G is the shear modulus, J is the torsional constant, and P is the pitch moment. The G values used for normalizing twist, CFRP00 (22 GPa) and CFRP30 (9 GPa), were estimated from the E values listed in Table 2 and are merely for comparison purposes. The CFRP00 hydrofoil shows a small positive twist angle, effectively increasing the incidence. Whilst the CFRP30 shows a negative twist angle, effectively decreasing the incidence angle. In both models, the twist angle has a linear dependence on incidence for any given Re$_c$. However, it can be observed that the magnitude of the twist angle varies significantly between the two models. On average, the CFRP30 experiences a twist 5.7 times larger than that of the CFRP00 under the same flow conditions. It can be seen that the dimensionless twist angle for each of these hydrofoils is nominally constant, indicating Re$_c$ independence and linear dependence on incidence.

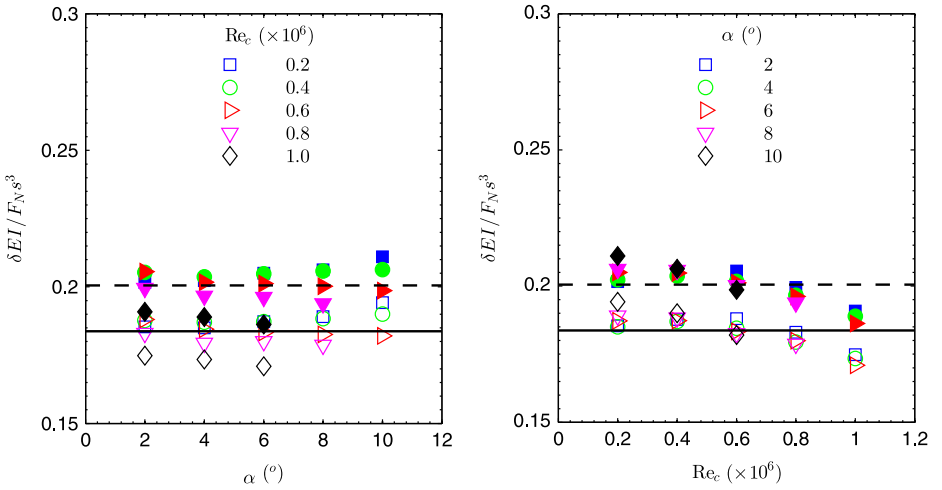


Fig. 21. Comparison of dimensionless leading edge (*open symbols*) and trailing edge (*closed symbols*) tip deflections for the Type II-CFRP30 hydrofoil against incidence (*left*) and Re_c (*right*). Mean leading edge dimensionless deflection, 0.183 (*solid lines*); mean trailing edge dimensionless deflection, 0.201 (*dashed lines*).

The experimental results show that hydrofoils made from isotropic material (SS and AL) have a similar hydrodynamic and hydroelastic response under static loading and pre-stall conditions. The same does not hold true for the composite hydrofoils. It is observed that the alignment of the carbon unidirectional layers has a significant effect on the hydrodynamic performance and hydroelastic response of the hydrofoils. The CFRP30 hydrofoil experiences an unloading caused by reduced incidence, as a result of the bend–twist coupling. It is evident from Fig. 10 that the higher the loading ($Re_c = 1.0 \times 10^6$ as compared to $Re_c = 0.2 \times 10^6$), the greater the reduction in lift, drag and pitching moment coefficients. The opposite is observed in the CFRP00 model, mainly in the lift and pitching moment coefficients. This is expected given that the bend–twist coupling is small for this case, as is evident from the lift spectra for this hydrofoil (see Fig. 15).

4. Summary and conclusions

The lift, drag, pitching moment, and hydroelastic deformations of flexible metal and composite hydrofoils, with two profiles, were investigated in a steady flow. The metal hydrofoils were manufactured from stainless steel and aluminum and the composites from carbon-fiber reinforced polymer. Two composite hydrofoils were tested, one with carbon unidirectional layers aligned along the spanwise axis of the foil, $\theta = 0^\circ$ (CFRP00), the other with the layers off-set with an angle $\theta = 30^\circ$ (CFRP30). All models had the same unswept trapezoidal planform but two section profiles were chosen – a standard NACA0009 and a modified NACA0009 profile with a thicker trailing edge. The modified profile was chosen for improved manufacturing of the composite hydrofoils. All hydrofoils were tested for a range of chord-based Reynolds number from 0.2×10^6 to 1.0×10^6 and within and incidence range of -15° to 15°.

Results show that the stainless steel and aluminum hydrofoils for both profiles have similar hydrodynamic and hydroelastic behavior. Minor differences are observed at low Reynolds numbers and in the magnitude of the dimensionless deformations. These differences may be attributable to the thicker trailing edge and the flow modification that this generates. Pre-stall forces were observed to be Reynolds number dependent for low values but became independent at 0.8×10^6 and greater. Forces and deflections were observed to be stable up to stall. Forces on all four metal hydrofoils compare closely for all pre-stall incidences and at the higher Reynolds numbers tested, showing these to be essentially independent of deformation. Dimensionless tip deflections were found to be independent of incidence and Reynolds number for the Type I geometry and mildly dependent on incidence and Reynolds number for the Type II. The results suggest that viscous effects, particularly at low and high incidence and low Reynolds number, have an effect on the hydrodynamic loading and hydroelastic response of the hydrofoils. Further work, focusing on the characterization of the flow around flexible hydrofoils and changes in the boundary layer regime, is required to provide a better insight into these phenomena.

The composite hydrofoils have a significantly different hydrodynamic and hydroelastic behavior, compared with the metal hydrofoils and with each other. The composite hydrofoils experienced much larger deflection and twist deformations under hydrodynamic loading. With the twist angle showing a dependence on the orientation of the carbon layers. The composite hydrofoil with layers oriented at $\theta = 0^\circ$ presented mild positive twist angle deformation, effectively increasing the incidence angle. Whilst the one with layers oriented at $\theta = 30^\circ$ presented negative and larger twist angles, effectively reducing the incidence. These deformations modify the hydrodynamic behavior of the hydrofoils by increasing the lift slope on the CFRP00 hydrofoil and reducing the lift slope in the CFRP30 hydrofoil with increasing Reynolds number. The CFRP30 hydrofoil also presented larger tip deflections with a strong Reynolds number dependence.

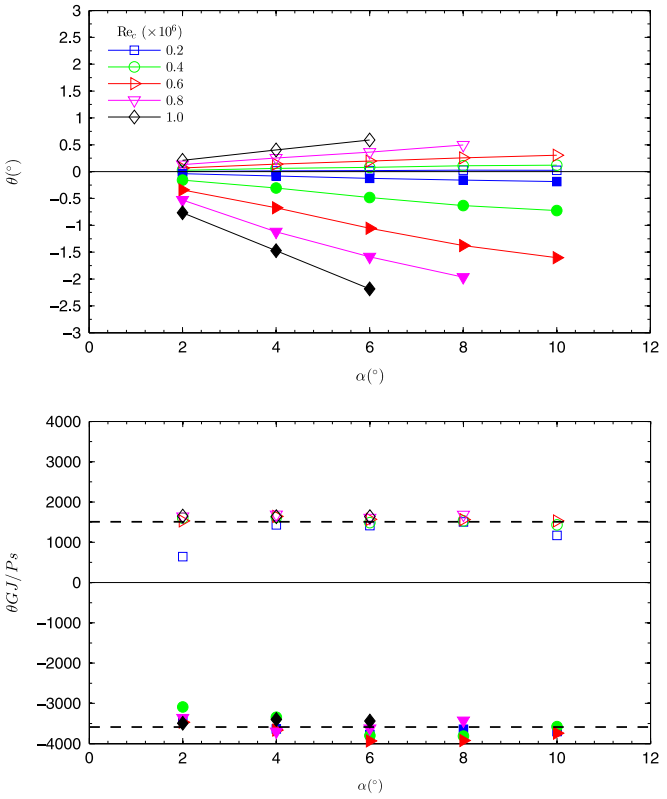


Fig. 22. Comparison of tip twist deflections for the Type II-CFRP00 (*open symbols*) and Type II-CFRP30 (*closed symbols*). The *top* panel shows the positive and negative twist angle of the CFRP00 and CFRP30 hydrofoils, respectively. The *bottom* panel shows the convergence of the dimensionless twist angle to approximately the same mean value (- -) for each hydrofoil.

For pre-stall incidences, none of the hydrofoils presented any significant vibration. For post-stall conditions, the response of the metal and composite hydrofoils was different. Both of the metal and the CFRP00 hydrofoils first mode frequencies (bending) in water are compared closely with those predicted from impact testing in air with an estimated added mass. Whilst the CFRP30 first mode was in between that of the aluminum and stainless steel hydrofoils and was not properly predicted from impact test results. The disagreement is attributed to the mixed deformation (bend–twist coupling) and the complex structural behavior due to the off-axis carbon layers under hydrodynamic loading.

The present work provides a high quality data set that will serve as a baseline for further testing under dynamic loading conditions and/or to validate numerical models concerned with the design, manufacture and testing of flexible hydrofoils and propellers. The results show that flexible hydrofoils have the potential for hydroelastic tailoring to suit specific applications such as enhanced hydrodynamic performance or reduced frequency related signatures for military vessels.

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